Functional & Performance Testing

Model Performance Test

Date	1 March 2026
Team ID	GACSE-114
Project Name	FitBuddy: AI Fitness Plan Generator Using Gemini Model
Maximum Marks	

1. Functional Testing Results

   The following table outlines the functional validation of the FitBuddy web interface and API endpoints.

Test Case ID	**Scenario (What to test)**	**Test Steps (How to test)**	**Expected Result**	**Actual Result**	**Pass/Fail**
TC-FN-01	Generate Plan via Web Form	Enter valid name, age, weight, goal, and intensity; submit form.	result.html displays 7-day routine; record stored in DB.	System generated personalized recommendations.	**Pass**
TC-FN-02	Input Validation (Web + Backend)	Submit invalid age (>120) or malformed JSON to API.	Backend rejects payload; API returns 422 with details.	System correctly validated user inputs.	**Pass**

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
TC-FN-03	Plan Update with Feedback	Submit feedback through result.html after generation.	Updated plan shown; updated_plan stored in DB.	AI generated updated personalized recommendations.	**Pass**
TC-FN-04	API Plan Generation	POST valid JSON payload to /api/v1/plans.	HTTP 201; Response includes user_id, plan, and timestamp.	API successfully returned AI-generated response.	**Pass**
TC-FN-05	API Feedback Update	POST user_id and feedback to /api/v1/plans/feedback.	Updated plan returned; DB record updated for matching user.	System updated existing records with new AI content.	**Pass**
TC-FN-06	Health Endpoint	Perform GET request to /health.	HTTP 200, body: {"status":"ok"}.	Service availability confirmed via health check.	**Pass**
TC-FN-07	AI Misconfiguration Handling	Trigger generation with GOOGLE_API_KEY missing.	Structured error response from AIServiceError.	System handled missing credentials with error message.	**Pass**

2. Performance Testing Results

The following table outlines the system's stability, latency, and resilience under load.

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
TC-PF-01	Plan Generation Latency	Run 30 concurrent requests; record p95 latency.	Should be under 3 seconds.	Average response time ~2.1 seconds.	**Pass**
TC-PF-02	Nutrition Tip Fast Path	Measure latency for quick nutrition tip generation.	No extra external AI call; tip pulled from local mapping.	Deterministic tips served instantly without API overhead.	**Pass**
TC-PF-03	AI Quota/Timeout Resilience	Simulate Gemini API quota exhaustion or timeouts.	Return 429 or 504 status codes; user-safe error message.	System handled external API limits without crashing.	**Pass**
TC-PF-04	DB Write Stability Under Burst	Send burst requests to /api/v1/plans.	Records persisted without corruption; rollbacks on failure.	Database maintained ACID compliance under concurrency.	**Pass**
TC-PF-05	Schema Compatibility	Start app on an old SQLite schema version.	App starts; generated_at backfilled for legacy rows.	Migration script successfully updated legacy data.	**Pass**

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
TC-PF-06	UI Responsiveness	Submit form and observe UI behavior during wait.	Loading overlay appears; prevents perceived freeze.	Overlay prevented double-submission and improved UX.	**Pass**

3. Conclusion

FitBuddy implements core functional and performance-oriented controls across input validation, API contract enforcement, error handling, database transaction safety, and basic observability. The AI layer includes fallback/error classification and response-time logging, while UI loading states improve perceived responsiveness during long-running operations.